

FULL NAME: _____

BLOCK: _____

L64 P1

Lesson 6-4 · Period 1 · Do Now

Name: _____ Period: _____ Date: _____

Do Now 1 — Translation identification: $h(x) = \ln(x + 2) - 1$

The logarithmic function $g(x) = \ln x$ is transformed to $h(x) = \ln(x + 2) - 1$. Which of the following are true? Select all that apply.

$g(x)$ is translated 2 units upward.

$g(x)$ is translated 2 units to the right.

$g(x)$ is translated 2 units to the left.

$g(x)$ is translated 1 unit downward.

$g(x)$ is translated 1 unit to the left.

The vertical asymptote shifts 2 units to the left.

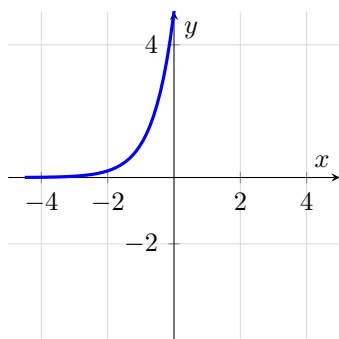
The vertical asymptote shifts 2 units to the right.

Do Now 2 — Graph $y = \log_4 x$; state key features

Graph the function $y = \log_4 x$ and identify the domain and range. List any intercepts or asymptotes. Describe the end behavior.

Do Now 3 — SAT/ACT: inverse of $f(x) = 5^{x+1}$

SAT/ACT The graph shows the exponential function $f(x) = 5^{x+1}$. Which of the following functions represents its inverse, $f^{-1}(x)$?



- Ⓐ $f^{-1}(x) = 1 + \log_5 x$ Ⓑ $f^{-1}(x) = \log_5 x - 1$
 Ⓒ $f^{-1}(x) = \log_5(x - 1)$ Ⓓ $f^{-1}(x) = \log_5(x + 1)$